Zhejiang University

Automation Curriculum

(Updated in April 2023)

1. Program Educational Objective

This program is an ideal choice for those who wish to pursue studies in Industrial Automation, System Control, and Robotics. It addresses latest technologies and their applications in control science and engineering, enabling students to explore new technology innovations to serve social demands.

2. Graduation Requirement

1) Curriculum Requirement

The curriculum leading to the degree of Bachelor of Science in Automation from ZJU requires 135 hours and is organized into required courses, technical electives, liberal education, Chinese language and society study and other electives.

3. Required Courses

3.1 Orientation and Professional Development

These courses introduce the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. They also provide the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Credit
ENG 100	Engineering Orientation	1

3.2 Foundational Mathematics and Science

These courses stress the basic mathematical and scientific principles upon which the engineering discipline is based.

Course Code	Course Name	2020	2021-
MATH 221	Calculus Ⅰ	4	4
MATH 231	Calculus Ⅱ	3	3
MATH 241	Calculus Ⅲ	4	4
MATH 257	Linear Algebra with Computational Application		3
MATH 285	Introduction to Differential Equation		3
MATH 286	Intro to Differential Equations Plus	4	
CHEM 102	General Chemistry Ⅰ	3	3
CHEM 103	General Chemistry Lab Ⅰ	1	1
PHYS 211	University Physics: Mechanics	4	4
PHYS 212	University Physics: Elec & Mag	4	4
PHYS 213	University Physics: Thermal Physics	2	2
PHYS 214	University Physics: Quantum Physics	2	2
	Total required	31	33

3.3 Technical Core

These courses stress fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of automation.

Course Code	Course Name	2020-
ECE 110	Intro to Electronics	3
ECE 120	Intro to Computing	4
ECE 220	Computer Systems & Programming	4
ECE 210	Analog Signal Processing	4
ECE 329	Fields and Waves Ⅰ	3
ECE 313	Probability with Engineering Application	3
ECE 340	Semiconductor Electronics	3
ECE 486	Control Systems	4

*ECE 498LFA**	*Lectures on Frontier Automation*	*1*
*ECE 498GRP**	*Graduation Research Project*	*8*
	Total required	37

**Instructed at Yuquan campus*

4. Elective Courses

4.1 Technical Electives

This elective requirement gives each student the freedom to develop a technical course of study in control science and technology of considerable breadth and focus. Students must complete a minimum of **22 credits** on technical electives.

Course Code	Course Name	Credits
CS 101	Introduction to Computing	3
CS 225	Data Structures	4
MATH213	Introduction to Discrete Mathematics	3
ECE 310	Digital Signal Processing	3
ECE 311	Digital Signal Processing Lab	1
ECE 314	Probability in Engineering Lab	1
ECE 342	Electronic Circuits	3
ECE 343	Electronic Circuits Laboratory	1
ECE 448	Artificial Intelligence	3
ECE 449	Machine Learning	3
ECE 470	Introduction to Robotics	4
*ECE 498RMC**	*Robot Modeling and Control*	*2.5*
*ECE 498 AR**	*Air-robots*	*2*
*ECE498 BDS**	*Big Data Security and Privacy Protection*	*1.5*
*ECE 498IBA**	*Introduction to Big data analytics and application*	*2*

**Instructed at Yuquan campus*

4.2 Liberal Education

The liberal education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the General Education requirements, students must take all courses for grade and complete courses based on the table below:

	2020-2021
1) **Humanities & Arts**	6
Two courses	
2) **Social & Behavioral Sciences**	6
Two courses	
3) **Culture Studies**	6
Western/Comparative	
Culture(s)	
One course	
Non-Western Culture(s)	
One course	
Total required	18

4.3 Free Electives

These unrestricted electives give the student the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students to define research specialties. Students are encouraged to take cross-discipline courses as free electives.

Free electives	2020	2021-2022	2023-
Total	5	4	3

5. ZJU Required Liberal Education

These courses introduce Chinese modern history, social development, government policies, etc., help students to improve their English and maintain a healthy lifestyle.

International students need to complete Chinese language study and Chinese society to fulfill the graduate requirement.

Course Code	Course Name	2020-2022 Credits	2023- Credits
CHIN 1001	Chinese Ⅰ	4	4
CHIN 1002	Chinese Ⅱ	5	4
CHIN 1003	Chinese Ⅲ	4	4
CHIN 1004	Chinese Ⅳ	4	4
CHIN1005	International Chinese Language Proficiency and Testing		2
CULT 2001	China Survey	3	3
	Total	20	21